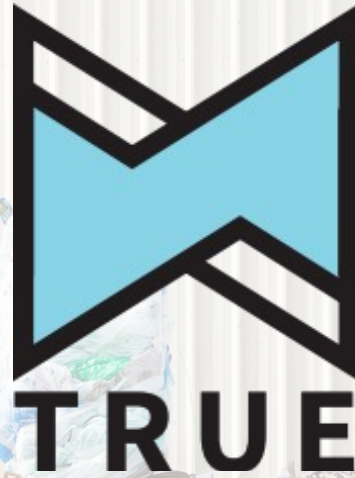




FACILITY CASE STUDY

**UC Berkeley
Gateway**

Berkeley, California, USA

Gateway

UC Berkeley

Certification level: Platinum

Diversion rate upon certification: 99.01%

Facility location: Berkeley, California, USA

Type of facility: University Building

Facility operations: Academic and administrative building for UC Berkeley's College of Computing, Data Science and Society, ~400,000 GSF

Website:

<https://facilities.berkeley.edu/true-certification>

Contact person and information: Brett Stuckey, Sustainability Manager, Turner Construction, and Audrey Pollack, Project Coordinator, Cal Zero Waste Gateway Project





Why Zero Waste?

- Gateway is UC Berkeley's first new college building in over 50 years, sitting along the highly visible Hearst Ave corridor. Pursuing TRUE for both Construction and Operations aligns with UC Berkeley's Zero Waste goals and its 2030 target to eliminate non-essential single-use plastics. An innovative research building of this scale successfully becoming zero waste also acts as a public statement to show what is possible, inspiring other projects to do the same.
- Key motivations: UC Berkeley's Zero Waste team initiated the pursuit in 2023; Turner and Ground Force (Turner's first TRUE project) quickly built a jobsite culture around it, proving ambitious outcomes don't require a cost premium, and building habits during construction that carry into Gateway's upcoming TRUE for Operations pursuit.



From Contract to Culture

- Simply writing zero waste requirements into contract documents doesn't guarantee they happen on site. Turning contract language into daily habits took ongoing meetings, presentations, and site walkthroughs to confirm that zero waste processes were being followed.
- Gateway involved a large number of people across UC Berkeley, Turner, and Ground Force in achieving TRUE certification. The hope is that the habits and lessons those individuals learned here will carry forward into whatever project they work on next.

Biggest Wins

- Gateway is Turner Construction's first TRUE project, achieving a 99.01% diversion rate including soil, recovering 66,751 tons of material, well above TRUE's 90% threshold.
- The team is especially proud of the project's source-separation strategy, estimated to have saved roughly \$1.8M versus a fully commingled waste strategy, proving zero waste strengthens the business case as well as the environmental one.
- The Gateway Zero Waste TGIF program was launched through a student-written grant that funded positions for UC Berkeley students to help pursue this certification, providing hands-on work experience and a dedicated team supporting the project.

Credit category: Redesign

- The Gateway team completed a full supply chain review by meeting with trade contractors during preconstruction, working with the curtain wall, mechanical, and electrical contractors to identify opportunities to eliminate waste before it reached the jobsite.
- As a result, the mechanical contractor implemented reusable transport containers for HVAC ductwork and a curtain wall contractor adopted a reusable bunk delivery system. This eliminated single-use timber crating and reduced wood, cardboard, and plastic waste across multiple trades.



Reusable transport containers used to eliminate single-use HVAC ductwork packaging.

Credit category: Reduce

- Timber pallet and crating waste was reduced through Permasteelisa's Hybrid Bunk delivery system, which replaced traditional single-use timber curtain wall crates with reusable aluminum and steel bunks. This saved an estimated 100+ tons of wood from the landfill.
- Turner also tracked waste generation in real time using Turner Tracker, a GreenHalo-based platform that compares projected material outcomes against actual diversion performance, helping the team catch and correct inefficiencies before they became waste.



Permasteelisa Hybrid Bunk reusable curtain wall delivery system.

Credit category: Reuse

- Inside the jobsite trailer, Turner implemented its Green Zone reusable-first standards. This included reusable dishware, silverware, and branded coffee cups to replace single-use disposables.
- When the Gateway site office was decommissioned, keyboards, whiteboards, TVs and chairs were transferred to two other Turner campus projects instead of being discarded, cutting new material costs for both projects.



Reusable dishware in the Turner project trailer.

Credit category: Compost (Re-Earth)

- A dedicated food waste collection bin was installed in the Turner trailer kitchen in February 2026. Collected food waste was processed through a Mill Food Recycler at Warren Hall, and a portion of the processed grounds was returned to Gateway for onsite trench composting.
- The remaining processed material supported UC Berkeley's ongoing vermicomposting research, evaluating decomposition using worms and varying mixtures of food grounds. This turned jobsite food waste into a live research partnership with campus.



Mill Food Recycler processing food waste from the Gateway jobsite trailer.

Credit category: Recycle

- Recyclable commodities were sent to specialized processors: scrap metal to Alco Iron & Metal for remelting into new steel, plastic film to SuperLink Plastic/Polyfit for new pellets, and concrete/asphalt to Argent Materials for recycled aggregate.
- When the team discovered wood waste was being incinerated instead of recycled, they immediately redirected it to Waste Management's Davis Street Transfer Station, a real-time correction that protected the project's diversion integrity and highest and best use of materials.

RECYCLABLE C&D MATERIALS		NON-RECYCLABLE C&D MATERIALS FOR DISPOSAL	
Wood: Untreated/painted Pallets, Wood Roofing, Dimensional Lumber & Crates	Inerts: Concrete, Asphalt, Bricks, Rocks, Gravel, Tile, Roofing Tiles & Other Aggregates	Asphalt Shingles	Painted or Stained Wood
Tree Debris: Trunks, Branches, & Trimmings	Fines: Dirt & Sand	Bagged Materials	Painted or Textured Drywall
Metal: Ductwork, Rebar, Other Ferrous & Non-Ferrous Metals	Consumer Recyclables: Aluminum, Plastic & Glass Bottles/Cans, Cardboard & Paper	Ceiling Tiles	Tar Roofing
		Film Plastics: Tarps, Shrink Wrap & Bags	Textiles: Roofing Felt, Carpet or Clothing
		Insulation: Fiberglass & Reflectix	
			

Davis Street waste management recyclables information.

Credit category: Zero Waste Reporting

- Gateway tracked diversion data for the full duration of construction and generated a climate impact report using the U.S. EPA WARM Model. Monthly financial data was compiled on disposal costs, recycling rebates, and avoided costs associated with the project's source-separation strategy.
- The financial analysis estimated the project saved roughly \$1.8M (including soil) compared to a conventional commingled disposal approach, demonstrating that rigorous zero waste reporting doubles as a business case for source separation.

Total Change in Energy Use (million BTU): (26,185.00)

This is equivalent to...		
Conserving	286	Households' Annual Energy Consumption
Conserving	4,507	Barrels of Oil
Conserving	217,389	Gallons of Gasoline

Total Change in GHG Emissions (MTCO₂E): (3,190.98)

This is equivalent to...		
Removing annual emissions from	677	Passenger Vehicles
Conserving	359,062	Gallons of Gasoline
Conserving	132,958	Cylinders of Propane Used for Home Barbeques
0.00018% Annual CO ₂ emissions from the U.S. transportation sector		
0.00018% Annual CO ₂ emissions from the U.S. electricity sector		

EPA WARM Model summary reports

Credit category: Diversion (Min. 90%)

- Gateway achieved a 99.01% diversion rate including soil, recovering more than 66,000 tons of construction material, well above TRUE's 90% minimum threshold.
- This performance reflects sustained source-separation discipline across the full duration of construction, from early sitework through the final, space-constrained months of the project when bin logistics became most challenging.



Source-separated bins on-site

Credit category: Zero Waste Purchasing

- Gateway follows Turner's Green Zone Environmentally Preferred Purchasing guidelines and the Waste Management Plan, prioritizing durable goods, sustainable paper/wood products, and salvaged or recycled-content furnishings, tracked through the LEED v4.1 Building Products Calculator.
- Additional practices included salvaging the concrete contractor's formwork, switching to bulk coffee with compostable filters, and joining Nespresso's aluminum pod take-back and recycling program.



Bulk coffee service and compostable filters in the project trailer.

Credit category: Leadership

- Gateway's Waste Management Plan set a project goal of greater than 90% diversion, led by Turner Sustainability Manager Brett Stuckey and reviewed monthly with Ground Force Enterprises and UC Berkeley Cal Zero Waste to track progress toward TRUE certification.
- Leadership also promoted Gateway's zero waste story externally, presenting at Greenbuild 2025 and encouraging the College of Computing, Data Science, and Society leadership to share the project's sustainability approach with peer institutions.



Monthly leadership review slideshow of diversion performance.

Credit category: Training

- The Waste Management Plan was a contract document for every trade contractor, reinforced by bilingual (English/Spanish) bin signage. Turner staff completed zero waste training via a first-day presentation, and purchasing agents were trained to embed diversion requirements into subcontracts.
- Turner labor crews were trained so thoroughly on proper sorting that they became informal “trash police” on site, reinforcing correct behavior alongside Ground Force's weekly bin audits.



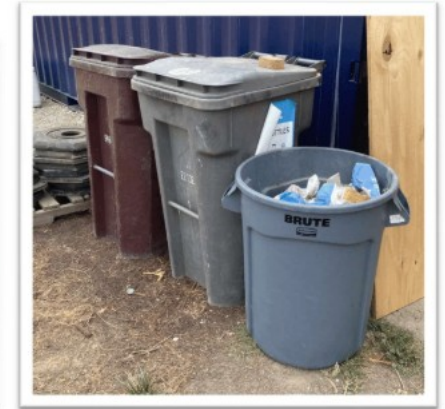
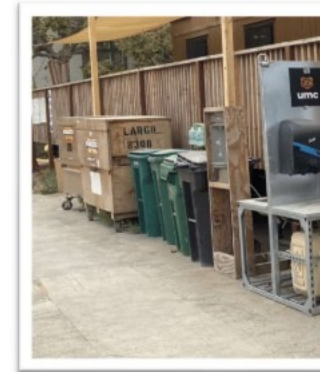
Bilingual signage on jobsite bins.

Credit category: Zero Waste Analysis

- Rather than a single audit, Gateway's waste analysis was continuous: Ground Force's on-site representative conducted weekly physical bin inspections and stayed in daily communication with Turner's team, feeding into monthly reviews of tonnage, diversion trends, and contamination levels (estimated at 2–5%).
- Findings drove real-time corrective action, including bin relocation, signage refreshes, and on-site worker education. This helped sustain the project's diversion performance through the end of construction.

Narrative

Still some contamination within the municipal solid waste bins (although a big improvement from previous visit!)



Recommendations:

- Lids on all bins
- Signs on top of lids with photos + Spanish
- Break areas
- Train laborers on how to pick up and dispose it
- Confirm with waste hauler it is going to a dedicated waste hauler
- Offer the bottles/cans to laborers to take home

Weekly bin audit conducted by Ground Force on the Gateway jobsite.

Credit category: Upstream Management

- Turner worked directly with trade contractors to reduce packaging waste at the source. Gateway's curtain wall work is a clear example. The team could compare the Level 1 storefront contractor's single-use wood crating with the Levels 2-5 Permasteelisa reusable metal bunkers to show the waste reduction made possible with reusables.
- Vendor accountability was also reinforced by UC Berkeley's Single-Use Plastic phase-out policy, which explicitly requires third-party vendors and contractors serving campus to eliminate non-essential plastics and provide sustainable alternatives.



Reusable metal delivery bunkers.

Credit category: Hazardous Waste Prevention

- Turner's safety program required Safety Data Sheets to accompany hazardous materials on-site, with licensed abatement contractors, never Turner's own crews, handling any physical abatement. Hazardous material manifests were retained for up to three years as well.
- Universal waste like batteries were collected and diverted through local e-waste recyclers, and the project's execution specifications explicitly restricted introducing hazardous materials into the finished building.



Labeled hazardous materials storage on the jobsite.

Credit category: Closed Loop System

- A Melaleuca tree, Coast Live Oak, and Canary Island Pine felled to enable Gateway's construction weren't sent to landfill. Bay Area Redwood milled, cured, and stored the wood for over three years before beginning fabrication.
- The result: 20+ Coast Live Oak and Melaleuca table tops and seven exterior picnic tables made from Gateway's own site trees, brought back onto campus to close the loop between construction and the building's finished interior and grounds.



Finished tables milled from trees felled during Gateway's site preparation.

Credit category: Innovation

- The Gateway Zero Waste TGIF program gave UC Berkeley students hands-on TRUE certification experience. They conducted site walkthroughs, launched composting initiatives, presented Gateway's story at the Greenbuild 2025 Conference, and supported the application.
- The Turner jobsite trailer was intentionally designed as a behavioral zero waste environment, making reusable dishware, bulk coffee, and compostable options the default choice. Turner's company-wide commitment to cut waste 95% by 2035 keeps the momentum going beyond this one project.



UC Berkeley TGIF students presenting Gateway's zero waste program.



Challenges and Lessons Learned

- MSW management deserved more attention. Consistent training, messaging, and incentives could have driven better recycling and composting participation. The final 2-3 months of the project also saw diversion rates dip due to limited bin space. Future projects should plan end-of-project logistics and bin zoning earlier.
- Looking back, the team would start collecting end-market documentation earlier, communicate documentation requirements to haulers and processors before construction begins, and maintain ticket-level records from day one.



Replicability & What's Next

- Gateway's construction-phase habits set up its upcoming pursuit of TRUE for Operations. Advice to offer for other campuses is to build zero waste requirements into the Waste Management Plan as a contract document and designate a dedicated lead to hold the project accountable.
- Gateway is meant as a learning model for future UC Berkeley construction. We identified what worked and hope to use this project as a repeatable template instead of starting from scratch. We also can use the financial results to show how source separation strengthens the business case.



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“Following the protocols set out by the TRUE Certification for Construction allowed UCB Gateway to produce incredible zero waste outcomes while reducing the cost compared to a business-as-usual approach. The comprehensive metrics and targets defined in the TRUE credits helped our project team think beyond recycling into a large variety of zero waste practices that had a profound impact on this landmark project. In addition, it has informed future waste practices for our projects as we know that zero waste initiatives can absolutely save a project money.”

*Brett Stuckey, Turner Regional
Sustainability Manager,
Northern California*



Zero waste is our mission
true.gbci.org